

Problem Set 2

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Problem 1. Outlining Proofs(a),(b) State Machine: 1. The set of state is $Q = \{\text{All possible permutations of } \{0, 1, 2, \dots, n-1\}\}$ **Start state:** Any one of the permutations of $\{0, 1, 2, \dots, n-1\}$.**Transition:**

$$(a_0, \dots, a_i, a_{i+1} \dots a_{j-1}, a_j, a_{j+1}, \dots, a_{n-1}) \rightarrow (a_0, \dots, a_j, a_{j-1} \dots a_{i+1}, a_i, a_{j+1}, \dots, a_{n-1})$$

If (i, j) is an inverted pair (i.e., $i < j$ but $a_i > a_j$).**Fianl state:** $q_f = (a_0, a_1, \dots, a_{n-1}) \mid (a_0 < a_1 < \dots < a_{n-1})$ which also implies that the list is sorted.(c) Derived variable: For any state $q = (a_0, a_1, \dots, a_{n-1})$, define

$$X(q) := a_0 \| a_1 \| \dots \| a_{n-2} \| a_{n-1}.$$

where $\|$ is concatenation operator.

Also $X(q)$ has only nonnegative integer value beacause, each term of $X(q)$ is part of state q. By definition each member of every valid state is defined over natural numbers, so number obtained from concatenating natural number must also be natural.

We know that digits on more right are more significant then terms on right to them and during transition we swap terms a_i and a_j where $i < j$ and $a_i > a_j$ so after swap value of left most term taking part in swap which is on positon i will decrease resulting in overall value of the final concatenated value decreasing. Meaning value of $X(q)$ will strictly decrease by each transition.

(d): For derived variable $X(q)$ from part (c) we know that it's value is defined over natural numbere and $X(q)$ is strictly decreasing.

Then by termination theorem we can conclude that state machine will terminate and from part(b) we know that final state will be fully sorted list.

Problem 2. A Fibonacci Sum

(a)

$$S = \sum_{n=1}^{\infty} \frac{F_n}{c^n}$$

$$S = \frac{F_1}{c} + \frac{F_2}{c^2} + \frac{F_3}{c^3} + \frac{F_4}{c^4}$$

$$\frac{S}{c} = \frac{F_1}{c^2} + \frac{F_2}{c^3} + \frac{F_3}{c^4}$$

$$S - \frac{S}{c} = \frac{F_1}{c} + \frac{F_2 - F_1}{c^2} + \frac{F_3 - F_2}{c^3} + \dots$$

$$S - \frac{S}{c} = \frac{F_1}{c} + \frac{F_0}{c^2} + \frac{F_1}{c^3} + \frac{F_2}{c^4} + \dots \quad [\because F(n) - F(n-1) = F(n-2)]$$

$$S - \frac{S}{c} = \frac{F_1}{c} + \frac{F_0}{c^2} + \frac{1}{c^2} \left[\frac{F_1}{c} + \frac{F_2}{c^2} + \frac{F_3}{c^3} + \dots \right]$$

$$S - \frac{S}{c} = \frac{1}{c} + \frac{S}{c^2} \quad [\because F_1 = 1, F_0 = 0]$$

$$\boxed{S = \frac{c}{c^2 - c - 1}}$$

(b) By calculating the first 50 digits i got,

0.00001000010000200003000050000800013000210003400055

Which suprisigly shows magically shows Fibonacci series terms in order every 5 digits.

Problem 3. Integral Method